Mediants and Collatz: Constructing Generalised Collatz Trajectories via Rational Approximation

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Abstract

We describe an application of the mediant tree algorithm of [2] to the construction of generalised Collatz trajectories. Given a generalised Collatz system (g, h) with odd step $x \mapsto gx + q$ and even step $x \mapsto x/h$ (where $h \nmid q$), a parity sequence realising a good rational approximant to $\theta = \log_h g$ is produced directly by the mediant tree as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$. We show how to convert such a mediant word into a concrete starting integer x_0 via a standard modular-arithmetic pipeline, and explain why the canonical greedy parity-sequence algorithm produces exactly the Sturmian word for slope $1/(1 + \log_h g)$. The parity–residue correspondence used in the pipeline is classical (Terras [3], Lagarias [4]); what is new is the *generative* role of the mediant tree as a transducer: it constructs the canonical balanced parity word and the rational approximant it realises in $O(\log N)$ iterations, each requiring $O(1)$ oracle evaluations (exact comparisons of g^o with h^e). We compare with Eliahou's continued-fraction bounding method [5], which provides lower bounds on Collatz cycle lengths via good approximants of $\log_2 3$ but gives no means of constructing a parity sequence realising any given approximant.

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1 Introduction

A *generalised Collatz sequence* is defined by a pair (g, h) with $g, h \geq 2$ and $\gcd(g, h) = 1$: an element x with $h \nmid x$ (an *odd* element) is mapped to $gx + q$ and an element with $h \mid x$ (an *even* element) is divided by h. The standard Collatz sequence corresponds to $g = 3$, $h = 2$, $q = 1$. The parity structure of a trajectory — the sequence of odd and even steps — is governed by $\theta = \log_h g$: a long-descending trajectory must have the ratio of even steps to odd steps approximate θ from above.

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The question this paper answers. A non-trivial cycle in a Collatz system must have its parity sequence balanced so that the trajectory closes: following the cycle once gives $h^e x_0 = g^o x_0 + qK$, i.e. $(h^e - g^o)x_0 = qK$. The product-form identity (see [5, 1]) gives $h^e = \prod_{j=0}^{o-1}(g + q/x_j)$, where $x_0, \ldots, x_{o-1}$ are the odd elements of the cycle. Taking the geometric mean of both sides, $h^{e/o}$ is the geometric mean of the numbers $g + q/x_j$. For $q > 0$ and positive cycle elements, the map $x \mapsto g + q/x$ is strictly decreasing, so its geometric mean equals $g + q/\hat{x}$ for a unique *mean estimate*

$$\hat{x} = \frac{q}{h^{e/o} - g},$$

which satisfies $x_{\min} \leq \hat{x} \leq x_{\max}$ (where $x_{\min}$ and $x_{\max}$ are the minimum and maximum odd elements) by monotonicity. In particular, $\hat{x}$ is an upper bound on $x_{\min}$: at least one odd cycle element must lie at or below the mean estimate. This quantity is large when $h^{e/o} \approx g$, i.e. when $e/o \approx \theta = \log_h g$. Since θ is irrational, $h^{e/o} \neq g$ for any rational e/o, but arbitrarily close rational approximants $e/o \approx \theta$ produce parity sequences where $\hat{x}$ is arbitrarily large. The mediant tree provides a canonical *generator* for such sequences: for any desired approximation quality, it produces an explicit e/o, the balanced parity word of length $e + o$ that realises it, and (via the pipeline of Section 4) a concrete starting integer x_0 whose trajectory follows that word exactly. This provides a systematic tool for exploring the structure of near-cycle parity sequences — the region of parity space where any non-trivial cycle, were it to exist, would have to live.

The companion paper [2] develops the *mediant tree*. The present paper describes the Collatz application that originally motivated it. The main contributions are:

1. A pipeline (Section 4) for converting a mediant word into a concrete starting integer x_0 whose generalised Collatz trajectory follows the prescribed parity pattern for exactly $e + o$ steps.
2. A proof (Section 5) that the canonical greedy parity-sequence algorithm produces exactly the Sturmian word for slope $1/(1 + \log_h g)$ — the same word the mediant tree generates.
3. A discussion (Section 7) situating the work relative to Terras [3], Lagarias [4], and Eliahou [5], and explaining what the generative transducer adds over prior descriptive approaches.

Organisation. Section 2 recalls the key definitions and results from [2]. Section 3 introduces generalised Collatz sequences and parity sequences. Section 4 presents the step-by-step pipeline from mediant word to Collatz trajectory. Section 5 describes the greedy algorithm and its Sturmian interpretation. Section 6 discusses why general Collatz trajectories are not Sturmian. Section 7 compares with Eliahou's approach. Section 8 describes observations on the carry polynomial's Laurent symmetry and future work directions: mutations of the limiting parity sequence and limiting parity sequences as a baseline for cycle analysis.

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2 Preliminaries

We recall the definitions and results from [2] that are used in this paper. All notation is consistent with that reference.

A *ratio* is a pair $(e, o) \in \mathbb{N}^2$ with $\gcd(e, o) = 1$, written e/o. Given a real $\theta > 1$, a ratio $r = e/o$ is a *good estimate* of θ if $|e - \theta o| < 1$. Two good estimates $r_{2d} = e_{2d}/o_{2d}$ and $r_{2d+1} = e_{2d+1}/o_{2d+1}$ form a *bounded pair* (r_{2d}, r_{2d+1}) if they bracket θ and satisfy $\Delta(r_{2d}, r_{2d+1}) := e_{2d+1}o_{2d} - e_{2d}o_{2d+1} = \pm 1$.

Given a bounded pair (r_{2d}, r_{2d+1}) and a weight $w \geq 0$, the *weighted mediant* is

$$\text{Mediant}(r_{2d}, r_{2d+1}, w) = \frac{e_{2d} + w\, e_{2d+1}}{o_{2d} + w\, o_{2d+1}}.$$

The *optimal integer weight* w^+ is the unique integer such that the pair

$$\big(\text{Mediant}(r_{2d}, r_{2d+1}, w^-),\ \text{Mediant}(r_{2d}, r_{2d+1}, w^+)\big)$$

brackets θ, where $w^- = w^+ - 1$; it is given in closed form by Theorem 8.1 of [2], which also guarantees $w^+ \geq 1$.

A *leaf* estimate has denominator $o = 1$; its mediant word is $\mathsf{E}^e\mathsf{O}$ (Definition 9.1 of [2]). The children of a bounded pair at the optimal weight w^+ themselves form a bounded pair (Corollary 8.2 of [2]).

The *mediant word* $\pi(r)$ of a ratio r produced by the mediant tree is a string over $\{\mathsf{O}, \mathsf{E}\}$ of length $e + o$ encoding the sequence of weighted-mediant choices made to reach r from the initial bounded pair. It is a lower Christoffel word (Lemma 9.3 of [2]), ends in O (Theorem 12.1 of [2]), is OO-free (§12 of [2]), begins with EE for every non-leaf estimate (Theorem 12.4 of [2]), and is identical to the Sturmian word for slope $1/(1 + \theta)$ at all positions except possibly the last (Theorem 12.6 of [2]).

2.1 Structural polynomials [6]

We import the following definitions and result from [6].

Let w be a parity sequence with $o \geq 1$ odd steps and e even steps. Reading w from left to right, maintain a running count of E-steps seen so far; κ_j is the value of that count at the moment the $(j+1)$-th O-step is encountered (i.e. the number of E's strictly before it). Concretely: κ_0 is the number of leading E's before the first O (so $\kappa_0 = 0$ iff w begins with O); each subsequent κ_j is κ_{j-1} plus the number of E's between the j-th and $(j+1)$-th O. The sequence $(\kappa_0, \ldots, \kappa_{o-1})$ is the *κ-tuple* of w (Definition 2 of [6]); it is a non-decreasing sequence of integers recording how the E-count accumulates across the O-steps of w. For an OO-free (deterministic) word, each consecutive pair of O's has at least one E between them, so the inequalities are strict:

$$0 \;\leq\; \kappa_0 \;<\; \kappa_1 \;<\; \cdots \;<\; \kappa_{o-1} \;<\; e + \kappa_0;$$

(when $\kappa_0 = 0$, i.e. the word begins with O, this recovers $\kappa_{o-1} < e$; when $\kappa_0 \geq 1$ the word ends in O and $\kappa_{o-1} = e$, which satisfies $\kappa_{o-1} < e + \kappa_0$ since $\kappa_0 \geq 1$). For a non-deterministic word (containing OO) adjacent O's contribute equal consecutive κ values, so the strict inequalities relax to $\leq$.

The *structural polynomial* of w (Definition 3 of [6]), written $K(g,h)$ and also called the *carry polynomial*, is

$$K(g,h) \;=\; \sum_{j=0}^{o-1} g^{o-1-j}\, h^{\kappa_j} \;\in\; \mathbb{Z}[g,h].$$

When $o = 0$ we set $K = 0$.

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Theorem 2.1 (Path identity, Theorem 1 of [6])**.** *Let (g,h,q) be a generalised Collatz basis and w a parity sequence with o odd steps and e even steps. If $x_n = \Phi_w(x_0)$ is the value obtained by applying w to x_0, then*

$$h^e\, x_n \;=\; g^o\, x_0 \;+\; q\, K(g,h).$$

2.2 Laurent polynomials and palindromic structure

A *Laurent polynomial* in variables g, h is a finite sum $\sum_{a,b} c_{ab}\, g^a h^b$ where the exponents a, b range over integers (possibly negative), with integer coefficients c_{ab}. Laurent polynomials form a ring $\mathbb{Z}[g^{\pm 1}, h^{\pm 1}]$ containing the ordinary polynomial ring $\mathbb{Z}[g,h]$ as a subring.

A Laurent polynomial $F \in \mathbb{Z}[g^{\pm 1}, h^{\pm 1}]$ is *self-reciprocal with centre (p,r)* if

$$F(g,h) \;=\; g^p h^r\, F(g^{-1}, h^{-1}).$$

Equivalently, after the substitution $g = G \cdot g_0$, $h = H \cdot h_0$ with $g_0^p h_0^r = 1$, the polynomial is invariant under $(G,H) \mapsto (G^{-1}, H^{-1})$: it is a Laurent polynomial in G, H symmetric under simultaneous inversion. (We use (p,r) rather than (p,q) to avoid collision with the Collatz offset q.)

Why palindromes produce self-reciprocal polynomials. Consider a polynomial $F(g,h) = \sum_{j=0}^{n-1} g^{a_j} h^{b_j}$ whose exponent pairs (a_j, b_j) are determined by the positions of a fixed letter (say O) in a word w. If w is a *palindrome* — i.e. w reads the same forwards and backwards — then reflecting w maps the j-th O from the left to the j-th O from the right, giving the pairing $(a_j, b_j) \leftrightarrow (a_{n-1-j}, b_{n-1-j})$. For the carry polynomial K of a palindromic word specifically, the reflection swaps

$$(o-1-j,\ \kappa_j) \longleftrightarrow (j,\ e-\kappa_j),$$

so that for each term $g^{o-1-j}h^{\kappa_j}$ in K there is a matching term $g^j h^{e-\kappa_j}$. Summing over all pairs gives $K(g,h) = g^{o-1}h^e\, K(g^{-1}, h^{-1})$, i.e. K is self-reciprocal with centre $(o-1, e)$.

For a mediant word $\sigma = \mathsf{E} \cdot P \cdot \mathsf{O}$ with P a palindrome, the leading E shifts every κ_j by 1 and the trailing O contributes the unpaired term h^e. Separating $C = K - h^e$, the palindrome mechanism applied to the interior O's yields $C(g,h) = g^o h^{e+1} C(g^{-1}, h^{-1})$: centre $(o, e+1)$.

This is the mechanism by which the palindrome theorem for mediant words (Theorem 12.5 of [2]) implies the Laurent symmetry of the carry polynomial $C = K - h^e$ observed in Section 8.1.

3 Generalised Collatz Sequences and Parity Sequences

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A *generalised Collatz sequence* uses an odd step $x \mapsto gx + q$ and an even step $x \mapsto x/h$, where $g, h \in \mathbb{N}$ with $g, h \geq 2$ and $\gcd(g,h) = 1$, and $q \in \mathbb{Z}$ with $q \neq 0$ and $h \nmid q$. The parity sequence structure depends only on g and h through $\theta = \log_h g$; the value of q affects only the concrete starting residue x_0 (Step 4 of the pipeline in Section 4). A *parity sequence* is a string over $\{\mathsf{O}, \mathsf{E}\}$ recording the sequence of odd and even steps along a trajectory; O denotes an odd step and E an even step.

For a trajectory to descend on average, the ratio of even steps to odd steps must exceed $\log_h g$: if a segment of length $e + o$ has e even steps and o odd steps, the net effect is multiplication by g^o/h^e, which is less than 1 iff $e/o > \log_h g$. The boundary case $e = \lceil o \log_h g \rceil$, o fixed, gives a good estimate e/o of $\log_h g$ (when $\gcd(e,o) = 1$), and corresponds to the upper-bound condition

$$h^{e-1} < g^o < h^e.$$

A parity sequence realising a good estimate e/o must therefore have exactly o copies of O and e copies of E. The mediant word generator produces OO-free words by construction, as does the Sturmian word for slope $\alpha < 1/2$ (equivalently $\theta > 1$). For $h = 2$, OO-freeness is a necessary condition for realisability as a deterministic Collatz trajectory: each odd step $x \mapsto 3x+1$ produces an even result (since $3x+1$ is even when x is odd), forcing at least one E before the next O; an OO in the parity sequence would require an odd element mapping directly to another odd element, which is impossible. Conversely, for $h = 2$, any OO-free word is realisable. The parity of x_i depends only on $x_0 \bmod 2^{\kappa_i+1}$, where κ_i is the number of E-steps before position i. Since the word is OO-free, consecutive O-steps are separated by at least one E-step, so the κ_i are strictly increasing at each O. Each of the $e+1$ levels $\kappa = 0, \ldots, e$ therefore contributes exactly one fresh bit, so the set of x_0 whose trajectory follows the word is exactly one residue class modulo 2^{e+1}. The modular inversion in Step 4

of the pipeline produces a valid x_0 in this class [6]. Thus for $h = 2$, a parity word is realisable as a deterministic Collatz trajectory if and only if it is OO-free.[1]

4 From Mediant Word to Collatz Trajectory: the Pipeline

The mediant tree applies to any real $\theta > 1$, not only to logarithms. The following pipeline converts a mediant word into a concrete Collatz starting integer for any (g, h) with $\theta = \log_h g$.

Step 1: Choose a real θ and an initial bounded pair. Select any real $\theta > 1$ and take the initial integer bounded pair the unique Farey-adjacent pair of integer rationals $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$ bracketing θ, ordered as (r_0, r_1) so that r_1 is the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). For the standard Collatz system take $\theta = \log_2 3$; for the generalised system (g, h) take $\theta = \log_h g$. The construction applies equally to any irrational θ (e.g. π, $\sqrt{2}$), in which case it produces rational approximants without a Collatz interpretation.

Step 2: Run the mediant tree. Apply the optimal weight formula (Theorem 8.1 of [2]) to produce a sequence of mediant words $\pi(r_{2d+1})$ of increasing length. At depth d, the word $\pi(r_{2d+1})$ has o_{2d+1} odd steps and e_{2d+1} even steps, and e_{2d+1}/o_{2d+1} is a good estimate of θ satisfying $h^{e-1} < g^o < h^e$ (upper bound) or $h^e < g^o < h^{e+1}$ (lower bound).[2]

Step 3: Derive the path identity. For a chosen mediant word $\sigma = \pi(r_{2d+1})$, form its κ-tuple and carry polynomial $K(g, h)$ as defined in Section 2.1. Since σ is OO-free the κ-tuple is strictly increasing. Theorem 2.1 then gives the *basic path identity*

$$h^e\, x_n \;=\; g^o\, x_0 + q\, K(g, h),$$

where x_0 is the starting value and x_n the value after $e + o$ steps following σ.

The *shifted path identity* follows immediately by substituting $x_0 \leftarrow x_0 + h^{e+1}t$ and $x_n \leftarrow x_n + g^o h\, t$ for any $t \in \mathbb{N}_0$:

$$h^e\big(x_n + g^o h\, t\big) \;=\; g^o\big(x_0 + h^{e+1}t\big) + q\, K(g, h),$$

which holds since the two new t-terms cancel ($h^e \cdot g^o h\, t = g^o \cdot h^{e+1}t$). Hence the set of valid starting values is the arithmetic progression $x_0 + h^{e+1}\mathbb{Z}$, with corresponding end values $x_n + g^o h\,\mathbb{Z}$ (see also §8.1 for further structure of K).

Step 4: Instantiate to concrete (g, h) and solve for x_0. The function `path_identity_params(σ, g, h)` substitutes the concrete multiplier and base, computes K, and solves

$$x_0 \;\equiv\; -q\, K \cdot (g^o)^{-1} \pmod{h^{e+1}}$$

for a starting residue $x_0 \in [0, h^{e+1})$. The modular inverse $(g^o)^{-1} \pmod{h^{e+1}}$ exists because $\gcd(g^o, h^{e+1}) = 1$, which follows from the system hypothesis $\gcd(g, h) = 1$: since g and h

[1] Non-deterministic $h = 2$ Collatz maps — where the odd/even instruction is supplied externally rather than derived from the state — do permit adjacent OO symbols; see [8].

[2] These inequalities follow directly from the good-estimate condition $|e - \theta o| < 1$ with $\theta = \log_h g$: an upper estimate has $e - 1 < \theta o < e$, giving $h^{e-1} < h^{\theta o} = g^o < h^e$; a lower estimate has $e < \theta o < e + 1$, giving $h^e < g^o < h^{e+1}$.

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share no prime factor, neither do any powers of g and h. x_0 is an element whose parity sequence follows σ for exactly $o+e$ steps and then continues with an E (an *even-continuing* residue). In total, $h-1$ residues mod h^{e+1} follow σ; the formula selects the unique even-continuing one, and the other $h-2$ continue with an O. (For $h=2$ uniqueness is immediate since $h-1=1$; the general count follows from the fact that since g^o is invertible mod h^{e+1}, exactly *one* residue mod h^{e+1} satisfies $h^{e+1} \mid g^o x_0 + qK$, and among the h residues $x_0, x_0+h^e, \ldots, x_0+(h-1)h^e$ that agree mod h^e, exactly one has $h \nmid x_n$, because the final O gives $x_n \equiv g x_{n-1} + q \pmod{h}$ and $gr+q \equiv 0 \pmod{h}$ has exactly one solution r since g is invertible and $h \nmid q$.)

Step 5: Read off the approximation data. The output tuple $(o,\ e,\ e/o,\ x_0)$ encodes:

- e/o — a rational approximant to θ, with $e/(o+e)$ and $o/(o+e)$ the even and odd step densities;
- x_0 — the even-continuing starting residue $(\bmod\ h^{e+1})$ whose $gx+q$ trajectory realises the mediant word σ for exactly $o+e$ steps.

Setting $t=0$ gives $x_n = (g^o x_0 + qK)/h^e$, which is also determined by x_0. As $d \to \infty$ the ratio e_{2d+1}/o_{2d+1} converges to θ, with convergence rate governed by the continued-fraction expansion of θ.

Remark. Evaluating the Collatz iteration from x_0 is *not* an efficient way to generate the parity sequence: it requires arithmetic with integers of size $O(h^e)$, which grows exponentially in the word length. The parity sequence itself is produced directly and efficiently by the mediant tree. At each depth one comparison of g^o with h^e is required; finding the optimal weight w^+ takes $O(\log w^+)$ comparisons at that depth, but $\sum_d \log w_d^+ = O(\log N)$ over all d, so the total real-arithmetic work to produce a prefix of length N is $O(\log N)$ comparisons of numbers of size $O(h^e)$. The value of x_0 lies in establishing that a concrete Collatz trajectory realising the sequence exists — not in using it to regenerate the sequence.

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Remark (residue constraint). To the extent that a $3x+1$ trajectory matches a factor of the Sturmian word for $\theta = \log_2 3$, every odd element of the trajectory is congruent to 1 or 3 modulo 8. This is a consequence of the Sturmian alphabet being $\{\mathsf{EO}, \mathsf{EEO}\}$: an odd step from a residue $r \equiv 1 \pmod 8$ produces $v_2(3r+1) = 2$ even steps (an EEO block), while $r \equiv 3 \pmod 8$ produces $v_2(3r+1) = 1$ even step (an EO block); residue 5 (mod 8) forces the parity sequence $\mathsf{OEEE}\cdots$, and residue 7 (mod 8) forces the parity sequence OEOEO; neither pattern occurs in the Sturmian word for $\log_2 3$, so neither residue can appear in a trajectory matching it. A full account is given in [7].

5 The Greedy Parity Sequence Algorithm

For any generalised Collatz system (g,h) with $g,h \geq 2$ and $\gcd(g,h)=1$, there is a simple greedy algorithm that generates the idealised parity sequence using only integer arithmetic — no floating-point evaluation of $\log_h g$ is required. Let $\mathit{short} = \lfloor \log_h g \rfloor$ and $\mathit{long} = \mathit{short}+1$ (the two possible E-run lengths, since $\log_h g \in (\mathit{short}, \mathit{long})$).

```
def collatz_log_h_of_g(g=3, h=2):
    e=o=0
    short=0
    while h**(short+1) <= g:
        short+=1
    long=short+1
    while True:
        if g**(o+1) < h**(e+short):
            yield from 'E' * short
            yield 'O'
            e+=short
            o+=1
        else:
            yield from 'E' * long
            yield 'O'
            e+=long
            o+=1
```

The condition $g^{o+1} < h^{e+short}$ is a pure integer comparison equivalent to $(o+1)\log_h g < e + short$, i.e. appending a *short* block $\mathsf{E}^{short}\mathsf{O}$ is sufficient to keep the ratio e/o above $\log_h g$; otherwise a *long* block $\mathsf{E}^{long}\mathsf{O}$ is needed. This is exactly the Beatty/Sturmian greedy rule: at each O step the algorithm selects the shorter run when the fractional part of $(o+1)\log_h g$ is already covered by the accumulated e, and the longer run otherwise.

The resulting sequence is the Sturmian word for slope $\alpha = 1/(1+\log_h g)$, characterised by $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ for all $n \geq 0$.

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Proof that the greedy output equals the Sturmian word. Write $\theta = \log_h g$ (irrational: if $\log_h g = a/b$ then $g^b = h^a$, impossible for coprime $g, h \geq 2$) and $s = \lfloor \theta \rfloor$. Let e_k denote the total number of E-steps output after k complete O-steps. We claim that $e_k = \lceil k\theta \rceil$ for all $k \geq 0$.

Base case. $e_0 = 0 = \lceil 0 \rceil$. ✓

Inductive step. Assume $e_k = \lceil k\theta \rceil$. Since $\lceil k\theta \rceil - 1 < k\theta \leq \lceil k\theta \rceil$ and $s \leq \theta < s+1$, we have

$$\lceil k\theta \rceil + s - 1 < (k+1)\theta < \lceil k\theta \rceil + s + 1.$$

The greedy appends a *short* block (s copies of E then O) iff $g^{k+1} < h^{e_k+s}$, equivalently $(k+1)\theta < e_k + s = \lceil k\theta \rceil + s$. In that case $\lceil (k+1)\theta \rceil = \lceil k\theta \rceil + s$, so $e_{k+1} = e_k + s = \lceil (k+1)\theta \rceil$. Otherwise $(k+1)\theta > \lceil k\theta \rceil + s$ (strictly, since θ is irrational), so $\lceil (k+1)\theta \rceil = \lceil k\theta \rceil + s + 1$, and $e_{k+1} = e_k + s + 1 = \lceil (k+1)\theta \rceil$. Either way $e_{k+1} = \lceil (k+1)\theta \rceil$, completing the induction.

Now the k-th O (1-indexed) is emitted at position $e_k + k - 1$ (0-indexed) in the output. With $e_k = \lceil k\theta \rceil$ this is position $\lceil k\theta \rceil + k - 1 = \lfloor k(1+\theta) \rfloor$. The Sturmian word has $s_n = \mathsf{O}$ at position n iff $n = \lfloor k(1+\theta) \rfloor$ for some $k \geq 1$, which matches exactly. Since a binary word over $\{\mathsf{O}, \mathsf{E}\}$ is uniquely determined by the index positions of its O's, the greedy output and the Sturmian word are identical. By Theorem 12.6 of [2], the mediant word $\pi(r_{2d+1})$ also equals the Sturmian word at all interior positions. Hence the greedy output, the Sturmian word, and the mediant word are all the same word (up to the final letter of the mediant word). □

For the standard case $g = 3$, $h = 2$ the first 60 symbols are:

EEOEEOEOEEOEOEEOEEOEOEEOEOEEOEEOEOEEOEOEEOEOEEOEEOEOEEOEOEEO,

with O-density converging to $1/(1 + \log_2 3) = \log_6 2 \approx 0.387$ — the *ideally balanced* parity pattern, descending as slowly as theory permits. For $g = 5$, $h = 2$ (*short* = 2, *long* = 3, since $\log_2 5 \approx 2.322$) the first 60 symbols are:

EEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEO,

with O-density converging to $1/(1 + \log_2 5) \approx 0.301$. The $5x + 1$ system has three known cycles; the mediant word describes the idealised parity pattern of a trajectory that descends at the asymptotically optimal rate for that system.

6 General Collatz Sequences are not Sturmian

The Sturmian word for slope $\alpha = 1/(1 + \theta)$ consists of exactly two block types: $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$, interleaved in the Sturmian pattern. These are precisely the leaf words of the mediant tree for θ. A Sturmian word can never contain a block of any other length.

A general Collatz trajectory is far less constrained: an actual orbit can produce E-runs of any length, interleaved freely. The mediant word (equivalently, the Sturmian word) is the *ideally balanced* special case.

Moreover, even among trajectories that approximate $\log_h g$ closely, the Sturmian alphabet $\{\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}, \mathsf{E}^{\lceil\theta\rceil}\mathsf{O}\}$ is not the only primitive alphabet that can arise. For example, the three known cycles of the $5x + 1$ system have alphabets $\{\mathsf{EO}, \mathsf{EEEEO}\}$, $\{\mathsf{EO}, \mathsf{EEEEEO}\}$, and $\{\mathsf{EO}, \mathsf{EEEO}\}$ — each containing the block EO of length 2, which does not appear in the Sturmian alphabet $\{\mathsf{EEO}, \mathsf{EEEO}\}$ at all. The mediant tree generates parity sequences over the Sturmian alphabet only; it produces the optimal approximation to θ but does not exhaust all primitive alphabets whose e/o ratio approximates θ.

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7 Relationship to Eliahou's Work

Classical context. The connection between parity sequences and rational approximation of $\log_h g$ is classical. Terras [3] introduced the parity sequence as a tool for analysing Collatz orbits and observed that a descending trajectory requires the ratio of even to odd steps to exceed $\log_2 3$. Lagarias [4] gave a systematic account of the parity–residue correspondence, relating the number of odd and even steps in a segment to the residue class of the starting integer modulo a power of h. Eliahou [5] sharpened this to explicit lower bounds on nontrivial cycle lengths, using best rational approximants of $\log_2 3$ obtained from the continued-fraction expansion.

These works are *descriptive*: they characterise what parity sequences must look like and what constraints a cycle must satisfy, but they do not provide a means of constructing a parity sequence achieving a given approximation quality.

The generative transducer. The contribution of the present work is a *generative* one. The mediant tree of [2] is a transducer: given a target $\theta = \log_h g$, it simultaneously constructs a rational approximant e/o to θ and the canonical balanced parity word realising it, in $O(\log N)$ iterations, each requiring $O(1)$ oracle evaluations (exact comparisons of g^o with h^e). The approximation quality is controlled entirely by the depth to which the tree is expanded. The mediant word is not just any parity word with the right ratio of E's to O's: it is the *canonical* one, identical to the Sturmian word for slope $1/(1+\theta)$ at all interior positions (Theorem 12.6 of [2]), and palindromic in its interior (Theorem 12.5 of [2]). It is the lower Christoffel word $\mathrm{Ch}^-(e, o)$ — the unique *lower Christoffel* word with e copies of E and o copies of O (Lemma 9.3 of [2]), equivalently the unique conjugate of the form $\mathsf{E} \cdot P \cdot \mathsf{O}$ with P a palindrome among all circularly balanced words with these letter counts.

Comparison with Eliahou. Our algorithm constructs approximants directly from bounded pairs via the optimal weight formula (Theorem 8.1 of [2]), without continued-fraction computation; the continued-fraction connection emerges as a consequence of the Sturmian identity (Section 10 of [2]). Eliahou's framework gives lower bounds on cycle lengths but provides no construction of a parity sequence realising any given approximant.

The present work generalises in both directions: the mediant tree applies to any real θ (not just $\log_2 3$), and the pipeline of Section 4 produces a concrete trajectory for any generalised Collatz system (g, h, q) — with the system parameters chosen freely to match the approximation target.

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8 Observations and Future Work

8.1 Laurent symmetry of the carry polynomial

The palindrome theorem for mediant words (Theorem 12.5 of [2]) has a direct algebraic consequence for the carry polynomial $K(g, h)$.

Recall the carry polynomial $K(g, h)$ and path identity from Section 2.1 (Theorem 2.1).

Every mediant word has the form $\sigma = \mathsf{E} \cdot P \cdot \mathsf{O}$ where P is a palindrome (Theorem 12.5 of [2]). The leading E shifts every κ_j up by 1; the trailing O (the last odd step, $j = o - 1$) has $\kappa_{o-1} = e$ (all even steps precede it), contributing the term $g^0 h^e = h^e$. Separating this unpaired term, write

$$K(g, h) = h^e + C(g, h),$$

where $C = \sum_{j=0}^{o-2} g^{o-1-j} h^{\kappa_j}$ absorbs the contributions from the interior O's. The palindrome symmetry of P forces C to be self-reciprocal with centre $(o, e+1)$ in the sense of Section 2.2:

$$C(g, h) \;=\; g^o h^{e+1}\, C(g^{-1}, h^{-1}),$$

so centering by $g^{o/2} h^{(e+1)/2}$ yields a Laurent polynomial in $g^{1/2}, h^{1/2}$ invariant under simultaneous inversion.

By the mechanism of Section 2.2, this means the cumulative additive contributions *accumulate from the start to the midpoint* in a pattern that is the exact mirror image of the *descent from the midpoint to the end*: the κ-multiset of C is invariant under $\kappa_j \mapsto e + 1 - \kappa_j$.

This symmetry holds for every word of the form $\mathsf{E} \cdot P \cdot \mathsf{O}$ with P a palindrome, of which mediant words are the canonical example.

The implications of this symmetry for the structure of K at different depths of the mediant tree — and for the cycle closure question — are left as future work.

8.2 Mutations of the limiting parity sequence

The mediant tree generates a specific *mediant word* $\pi(r_{2d+1})$ for a given approximation scale: a finite word that is Sturmian up to its last letter, with palindromic interior, and OO-free, approximating the infinite limiting Sturmian sequence. This provides a natural reference object whose properties are well understood. Any parity word with the same e/o ratio can be viewed as a *mutation* of this mediant word, and different mutations conserve different properties while altering others.

Two examples illustrate how mutations can differ in what they conserve:

1. **Block-length mutations.** Replacing a Sturmian block $\mathsf{E}^\ell\mathsf{O}$ with $\mathsf{E}^{\ell'}\mathsf{O}$ where $\ell' \notin \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$ introduces a block length outside the Sturmian alphabet. This changes $e - \theta o$ by the nonzero integer $\ell' - \ell$, shifting e/o away from θ for deep mediant words where $|e - \theta o| < \frac{1}{2}$, and alters every term of $K(g, h)$ from the modified block's O onward. It also changes the residue condition on x_0.

2. **Reordering mutations.** Permuting the blocks of a Sturmian word — for example, rearranging $\mathsf{EEO} \cdot \mathsf{EEO} \cdot \mathsf{EO} \cdot \mathsf{EEO} \cdot \mathsf{EO}$ (i.e. $\pi(8/5)$) as $\mathsf{EEO} \cdot \mathsf{EEO} \cdot \mathsf{EEO} \cdot \mathsf{EO} \cdot \mathsf{EO}$ — keeps the same letters and hence the same e/o ratio, and preserves the multiset of block lengths. What changes is the *order* of the blocks, which alters the run structure and the carry polynomial $K(g, h)$. For $\theta > 1$, every Sturmian block $\mathsf{E}^\ell\mathsf{O}$ has $\ell \geq 1$, so no reordering can create an OO adjacency. The reordered word above is unbalanced (it contains both OEOEO and EEOEE) and is not a factor of any Sturmian word.

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These two examples differ in what they conserve. Block-length mutations change $e - \theta o$ by a nonzero integer, typically shifting e/o away from θ for deep mediant words where $|e - \theta o| < \frac{1}{2}$. Reordering mutations keep e/o fixed but change the internal structure of $K(g, h)$ and generally break the palindrome invariant (though some permutations that maintain the E·P·O decomposition may preserve it). In each case the mediant word serves as a baseline: starting from the ideal word and applying controlled mutations, one can ask how properties such as the mean estimate $\hat{x}$, the min/max cycle element bounds, or the modular closure condition respond to each class of change. This is left as a direction for future investigation.

8.3 Limiting parity sequences and cycle constraints

At each denominator o_{2d+1} produced by the MT algorithm the mediant tree identifies the *mediant word* $\pi(r_{2d+1})$, a finite approximant of the infinite limiting Sturmian sequence over the block alphabet $\{\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}, \mathsf{E}^{\lceil\theta\rceil}\mathsf{O}\}$ whose e/o ratio converges to $\theta = \log_h g$. This mediant word is the algebraic reference object for the scale in question. Its carry polynomial $K(g, h)$ has the interior part $C(g, h) = K(g, h) - h^e$ with a symmetric Laurent structure (Section 8.1), a consequence of the palindromic interior of every mediant word (Theorem 12.5 of [2]).

Whether this structure imposes constraints on cycle existence — and if so, what those constraints are — is not claimed here in either direction. The purpose of this section is simply to motivate the study of limiting sequences as the natural algebraic baseline for cycle analysis at any scale.

Two observations connect the limiting sequence to any actual cycle candidate.

1. **Exact closure requires arithmetic balance.** A cycle satisfies $x_n = x_0$, which via the path identity $h^e x_n = g^o x_0 + qK(g,h)$ yields the exact integer equation

$$(h^e - g^o)\, x_0 = q\, K(g,h).$$

   Because $\theta = \log_h g$ is irrational, $h^e \neq g^o$ for any finite parity word with $o \geq 1$, so any cycle requires a precise arithmetic balance between the multiplicative mismatch $h^e - g^o$ and the carry polynomial K. The limiting sequence for the relevant scale sets the algebraic context for this balance.

2. **Eliahou's constraints bound the letter-frequency scale.** For the standard $3x+1$ system, Eliahou's analysis [5] of the product form of the cycle equation $h^e = g^o \prod_j (1 + q/(gx_j))$ shows that the e/o ratio of any non-trivial cycle satisfies

$$\log_2 3 \;<\; \frac{e}{o} \;\leq\; \log_2\!\Big(3 + \frac{1}{m}\Big),$$

   where m is the minimum cycle element. This means any cycle word must sit in a narrow interval *just above* $\theta = \log_2 3$, and hence its letter frequencies are close to those of the limiting Sturmian sequence for the same scale. (This specific bound is for the $3x+1$ system; analogous product-formula constraints exist for general (g,h,q).) Whether "close in frequency" forces structural proximity to a mediant word — and whether that proximity is algebraically significant — are open questions that the limiting-sequence framework is designed to address.

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The mediant tree thus provides a natural baseline for cycle analysis: for any parity word with a given e/o ratio, the corresponding limiting sequence describes what the "ideal" word at that scale looks like algebraically. How a cycle candidate relates to this ideal — and whether the Laurent symmetry of the limiting carry polynomial plays any role in ruling cycles in or out — is left as a direction for future investigation.

References

[1] J. Seymour, *Extracting Parameters from Generalized Rational Collatz Cycles*, `https://wildducktheories.github.io/collatz/papers/21-extract-parameters/extract-param` Paper 21 (2025).

[2] J. Seymour, *Mediant Trees for Rational Approximation and Sturmian Word Generation*, `https://github.com/wildducktheories/collatz`, Paper 76 (2026).

[3] R. Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), 241–252.

[4] J. C. Lagarias, *The* $3x + 1$ *problem and its generalizations*, American Mathematical Monthly **92** (1985), 3–23.

[5] S. Eliahou, *The* $3x + 1$ *problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), 45–56.

[6] J. Seymour, *The Structural Polynomial of a Parity Sequence*, `https://github.com/wildducktheories/collatz`, Paper 71 (2026).

[7] J. Seymour, *A Regular Expression Language for the Collatz Graph*, `https://github.com/wildducktheories/collatz`, Paper 33 (2026).

[8] J. Seymour, *Non-Deterministic Collatz Maps Preserve Non-Trivial* $3x + 1$ *Cycles*, unpublished manuscript (2026).

Disclosures

Use of Artificial Intelligence

Large language models played a substantial role in the development of this paper, and the author considers it important to describe that role honestly.

Review and gap identification. Claude (Anthropic) and Gemini (Google) were used as critical reviewers across multiple structured review rounds. Each model was presented with the manuscript and asked to identify mathematical gaps, proof obligations, notational inconsistencies, and expository weaknesses. The resulting issue lists directly drove successive revisions and are documented in the paper's revision history.

Theorem formulation. The boundary between the author identifying a result and an AI model identifying it is not always sharp. In several instances the need for a theorem or lemma was recognised during AI-assisted review rather than by the author independently. The author regards this as a genuine intellectual contribution by the models involved, not merely automated checking.

Proof development. On several occasions, Aristotle (the code name used for the AI reviewer in this project's review workflow) did not merely flag a conjecture as unproven but supplied the proof that elevated it to a theorem. Claude also drafted proofs and theorem statements at various points in the development; in some cases these drafts were adopted with revision, in others they were rejected or replaced. The author regards such contributions as substantive, and they are part of the honest account of this paper's development.

Exposition. Claude was used to assist with drafting, restructuring, and checking revised passages of mathematical exposition. No AI-generated text was incorporated without review and revision by the author.

Responsibility. The overall research programme, the framing of results, all decisions about what to accept, defer, or reject, and final responsibility for the correctness of the paper rest with the author. The author does not regard AI assistance as diminishing the paper's validity, but believes readers are entitled to a complete account of how it was produced.

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Lean 4 Verification Status

During peer review, a reviewer produced an independent Lean 4 formalisation of selected results from the companion paper (Paper 76), including the exponential denominator growth theorem and the weighted-mediant identity. The author is grateful for this contribution, which helped identify and correct several proof gaps during the review process.

The author has chosen not to publish the Lean 4 development at this stage. Accordingly, **this paper makes no claim of Lean 4 certification or machine-checked verification**. Every theorem, lemma, and proof in this paper stands solely on its hand-argued proof as written in the text. Readers should not infer formal verification status from any part of this manuscript. A complete Lean 4 formalisation of the results in this paper is left for future work.

Reviewer's Comments

The following items identify open proof obligations, gaps, and vulnerabilities in this manuscript. See Paper 76 for the full status key. Item numbers continue from Paper 76 (RC-1 through RC-16). **Deferred** items have been noted but action is intentionally postponed; reviewers should not expect Deferred items to be actioned in the next revision, and new comments solely about the content of Deferred items will not be actioned.

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ID	Title	Status
RC-7	§4 Step 3: $K(g,h)$ undefined; path identity unproved	Closed
RC-8	§5: Beatty-sequence identification without proof	Closed
RC-9	§8.1: Laurent symmetry introduced without motivation	Resolved
RC-10	§3: OO-free $\Leftrightarrow$ realisable unargued	Resolved
RC-11	§1: Mean-estimate formula without derivation	Resolved
RC-12	§4 Step 4: modular-inversion hypotheses silent	Closed
RC-17	Heuristic dynamical bridge: tree weights to trajectory lengths	Deferred
RC-18	Density and measure-zero exceptions unaddressed	Deferred
RC-19	Modular residue transport degeneracies at high moduli	Deferred
RC-20	Computational scalability limits above 2×10^{10}	Deferred
N6	§2.1: chain $\kappa_{o-1} < e$ tightened to $\kappa_{o-1} < e + \kappa_0$	Resolved
N7	§2.2 / §8.1: symmetry centres corrected; q clash resolved	Resolved
N8	§3: sufficiency argument for OO-free $\Rightarrow$ realisable is a non-sequitur	Resolved
N9	§1 / RC-11: mean-estimate derivation misstated; $\hat{x}$ is not the geometric mean	Resolved
N10	Step 4: "exactly h^e residues satisfy ..." is wrong	Resolved
N11	Remark after Step 5: $O(1)$ comparisons claim incorrect	Resolved
N12	RC-7 and RC-9 body text still describes the problem, not the resolution	Resolved
N14	RC-17–RC-20 refer to content not in the manuscript; noted and deferred	Resolved

RC-7. §4 Step 3: $K(g,h)$ undefined; shifted path identity unproved. *Source: claude — Since: v0.14 — Status: Resolved*

$K(g,h)$ is now defined inline in §2.1 and the path identity $h^e x_n = g^o x_0 + qK(g,h)$ is stated and proved as Theorem 2.1. Step 3 now cites these directly.

RC-8. §5: Beatty-sequence identification without proof. *Source: claude — Since: v0.14 — Status: Resolved*

The proof in §5 already showed greedy output = Sturmian word by induction on $e_k = \lceil k\theta \rceil$. A closing sentence now explicitly closes the triangle: Theorem 12.6 of [2] gives mediant word = Sturmian word at interior positions, so all three coincide.

RC-9. §8.1: Laurent symmetry introduced without motivation. *Source: claude — Since: v0.14 — Status: Resolved*

§2.2 now motivates the Laurent symmetry via the palindrome structure of mediant words and establishes the correct identities: $K(g,h) = g^{o-1}h^e K(g^{-1},h^{-1})$ for a palindromic word (centre $(o-1,e)$), and $C(g,h) = g^o h^{e+1} C(g^{-1},h^{-1})$ for the carry polynomial of a mediant word $\sigma = \mathsf{E}\cdot P\cdot \mathsf{O}$ (centre $(o,e+1)$). Both centres have been corrected from the previously printed values (N7).

RC-10. §3: OO-free ⇔ realisable unargued. *Source: claude — Since: v0.14 — Status: Resolved*

The necessity direction ($\Rightarrow$) is correct for $h = 2$. The sufficiency argument ($\Leftarrow$) is a non-sequitur: "the strict κ-tuple condition ensures $\gcd(g^o, h^{e+1}) = 1$" is false — coprimality follows from $\gcd(g,h) = 1$ alone, regardless of the word, and coprimality does not by itself show that the solution follows the word. A correct argument for $h = 2$: the parity of x_i depends only on $x_0 \bmod 2^{\kappa_i+1}$; each of the $e+1$ levels contributes exactly one fresh bit, so the followers of an OO-free word form one residue class mod 2^{e+1} (pre-Round-7 Lean: `noOO_realisable`, `follows_h2_iff`). Fix: replace the κ-tuple coprimality claim with this bit-level argument.

RC-11. §1: Mean-estimate formula without derivation. *Source: claude — Since: v0.14 — Status: Resolved*

The derivation is misstated in two ways. First, the derivation "from the cycle closure identity $(h^e - g^o)x_0 = qK$, taking the geometric mean…" is a non-sequitur: the bracket $x_{\min} \le \hat{x} \le x_{\max}$ comes from the product-form identity $h^e = \prod_j (g + q/x_j)$, not from the K-identity. Second, the RC-11 body text claims $\hat{x}$ *is* the geometric mean of the odd cycle elements: this is false. For the $5x+1$ cycle through 13 (odd elements 13, 33, 83; $e = 7$, $o = 3$), $\hat{x} \approx 25.2$ but the geometric mean is ≈ 32.9 (Lean: `mean_estimate_not_geometric_mean`). Fix: replace the derivation sketch with: $h^{e/o}$ is the geometric mean of the numbers $g + q/x_j$ (from the product identity); since $x \mapsto g + q/x$ is monotone for $q > 0$ and positive elements, the bracket follows. State hypotheses $q > 0$ and positive cycle elements.

RC-12. §4 Step 4: modular-inversion hypotheses silent. *Source: claude — Since: v0.14 — Status: Resolved*

Step 4 now states explicitly that $(g^o)^{-1} \pmod{h^{e+1}}$ exists because $\gcd(g,h) = 1$ implies $\gcd(g^o, h^{e+1}) = 1$, citing the system hypothesis.

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RC-17. Heuristic dynamical bridge: tree weights to trajectory lengths. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper makes no claim about connecting tree depths to Collatz stopping times or trajectory lengths. The phrases "probabilistic or deterministic argument" and "trajectory lengths" do not correspond to any claim in the manuscript.

RC-18. Density and measure-zero exceptions unaddressed. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper makes no asymptotic claims about global trajectory behaviour or natural density of exceptional sets. The concepts of "measure-zero exceptions" and "natural density" are not present in the manuscript.

RC-19. Modular residue transport degeneracies at high moduli. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper contains no analysis of residue transport modulo 5 or modulo 6, nor any "local-to-global" assumptions at higher composite moduli. These concepts appear to have been imported from a different paper.

RC-20. Computational scalability limits above 2×10^{10}. *Source: gemini — Since: v0.14 — Status: Deferred*

Note (N14): the paper contains no computational benchmarks and no "Bridging Machine architecture". This comment is not directed at the present manuscript.

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N6. §2.1: chain $\kappa_{o-1} < e$ false for words ending in O. *Source: aristotle — Since: v0.14 — Status: Resolved*

The bound has been tightened to $\kappa_{o-1} < e + \kappa_0$. When $\kappa_0 = 0$ (word begins with O) this recovers the original $\kappa_{o-1} < e$. When $\kappa_0 \geq 1$ (word ends in O, so $\kappa_{o-1} = e$) the bound $\kappa_{o-1} < e + \kappa_0$ is satisfied since $\kappa_0 \geq 1$. The strict inequalities between consecutive κ_j are unchanged.

N7. §2.2 / §8.1: symmetry centres wrong; double use of q. *Source: aristotle — Since: v0.14 — Status: Resolved*

All three issues resolved: (a) §2.2 now states the correct palindromic identity $K(g,h) = g^{o-1}h^eK(g^{-1},h^{-1})$, centre $(o-1,e)$. (b) The carry polynomial identity now reads $C(g,h) = g^oh^{e+1}C(g^{-1},h^{-1})$, centre $(o,e+1)$. (c) The self-reciprocal centre is now written (p,r) rather than (p,q), with an explicit note that q is reserved for the Collatz offset.

N8. §3: sufficiency direction of OO-free $\Rightarrow$ realisable is a non-sequitur. *Source: aristotle — Since: v0.14 — Status: Resolved*

See RC-10 (Resolved) above.

N9. §1 mean-estimate derivation misstated; $\hat{x}$ is not the geometric mean. *Source: aristotle — Since: v0.14 — Status: Resolved*

See RC-11 (Resolved) above.

N10. Step 4: "exactly h^e residues" is wrong. *Source: aristotle — Since: v0.14 — Status: Resolved*

The parenthetical in Step 4 now correctly states that exactly *one* residue mod h^{e+1} satisfies $h^{e+1} \mid g^o x_0 + qK$ (since g^o is invertible), and explains the $h-1$ follower count via the final O-step congruence $gr + q \equiv 0 \pmod{h}$.

N11. Remark after Step 5: $O(1)$ comparisons claim incorrect. *Source: aristotle — Since: v0.14 — Status: Resolved*

The Remark now states that the total real-arithmetic work is $O(\log N)$ comparisons of numbers of size $O(h^e)$, with the $O(\log N)$ aggregate following from $\sum_d \log w_d^+ = O(\log N)$.

N12. RC-7 and RC-9 body text describes the problem, not the resolution. *Source: aristotle — Since: v0.14 — Status: Resolved*

Both bodies have been rewritten to describe the resolution: RC-7 now states that $K(g,h)$ is defined in §2.1 and the path identity proved as a theorem, with Step 3 citing both. RC-9 now states the correct palindrome and carry-polynomial identities with their centres, noting that both were corrected from the previously printed values.

N14. RC-17–RC-20 refer to content not present in the manuscript. *Source: aristotle — Since: v0.14 — Status: Resolved*

RC-17–RC-20 have been dismissed. Each cites concepts absent from this manuscript: trajectory lengths and stopping-time dynamics (RC-17), natural density of exceptional sets (RC-18), residue transport modulo 5 and 6 (RC-19), and the "Bridging Machine architecture" (RC-20). The comments appear to have been generated against a different paper or a conflated description of multiple papers.

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